



## DS 412: Matrix Computations for Data Science

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| Course Code   | DS 412                                                                                                                                                                                             |
| Course Name   | Matrix Computations for Data Science                                                                                                                                                               |
| L-T-P-C       | 3-0-2-4                                                                                                                                                                                            |
| Prerequisites | IC 110-Engineering Mathematics, IC 111-Linear Algebra; From 2022 UG Batches                                                                                                                        |
| Intended for  | Discipline core for B. Tech. Data Science and Engineering, Discipline elective for B. Tech. Computer Science and Engineering, Electrical Engineering, Free elective for other B. Tech. disciplines |
| Approval      | 54th BoA (May 2024)                                                                                                                                                                                |

### Course Contents

- **Basics:** Errors in numerical computations. Review matrices and transformations, Matrix and Vector Norms. (2 lectures)
- **Matrix factorizations:** Cholesky factorization, QR factorization, Householder reflectors and Givens rotations. LU factorization and Gaussian elimination. Numerical stability Pivoting strategies and solution of triangular and full systems by factorization. (5 lectures)
- **Eigenvalue problem:** Basic theory, Power method, bisection method, QR algorithm. Similarity reduction. (4 lectures)
- **Linear least squares problem:** Singular value decomposition, Moore-Penrose Pseudoinverse. Perturbation theory. Least square and normal equations. SVD and rank deficiency, Principal Component Analysis, Linear Discriminant Analysis. (6 lectures)
- **Iterative methods for linear systems :** Iterative methods for linear systems Iterative methods: Jacobi, Gauss-Seidel and SOR iterations. Kronecker product. Krylov subspace methods, conjugate gradient method, preconditioning. (5 lectures)
- **Sparse and banded linear systems:** Storage schemes for banded and sparse matrices, Sparse matrices and sparse solutions: approximate inverses, eigenvalues, incomplete factorization. Matrix regularization: matrix completion. (6 lectures)

### Laboratory Modules

Lab would be conducted on a 2-hour weekly slot. Lab sessions would be conducted in tandem with the theory course so the topics for problems given in the lab are already initiated in the theory class.

### Text Books

1. Trefethen, Lloyd N., and David Bau III, **Numerical linear algebra**, Vol. 50, SIAM, 1997.
2. Eldén, Lars., **Matrix methods in data mining and pattern recognition**, Vol. 4. SIAM, 2007.

### References

1. Watkins, David S., **Fundamentals of matrix computations**, Vol. 64, John Wiley & Sons, 2004.
2. Demmel, James W., **Applied numerical linear algebra**, Vol. 56, SIAM, 1997.
3. Golub, Gene H., and Charles F. Van Loan., **Matrix computations**, Vol. 3, JHU press, 2012.
4. Cullen, Charles G., **An introduction to numerical linear algebra**, PWS Publishing Company, 1994.